

PROJECTOR: ₹25 TO 200



This is a quick and dirty build, and hardly requires any time. This allows you to use your smartphone as a projector, and all it really requires is a magnifying glass.

Aditya Madanapalle | aditya@digit.in

MATERIALS NEEDED

- Magnifying Glass: ₹25 to 200
- Cardboard: Free



You can build this in under ten minutes by using an existing box, but we recommend taking the trouble to make your own box, or use parts of the box to make a smaller box.

STEP 01 Go to a stationery store and buy the biggest magnifying glass that you can find. If it offers 5x magnification, that is great. The bigger the glass, and the more it magnifies, the larger will be your projection. The distance of projection will also increase.

STEP 02 Go to a dark room, and roughly check out the distance to which your magnifying glass can project the screen of the phone. This will give an idea of how large a box you will need. Although you can start off with a box, it is a good idea to build your own one, considering that the size of the box needed depends on the size of the phone and the magnifying glass.

STEP 04 Measure the length and breadth of the phone, and leave at least about 0.3 mm on all sides. The depth of the box depends on the distance that you checked in the dark room. Finally, if the magnifying glass is bigger than the phone, then the size of the box has to be increased to accommodate that. If using an existing box, avoid ones that open in the front, as against one that opens at the top. This is because you will have to reach into the box to adjust, which is painful, especially with touchscreens.



STEP 03 Trace out the shape of the magnifying glass on one side of a cardboard, or a flap of a box. Cut through this circle. There is no need to remove the handle from the glass. In fact, the handle can be used to tape the glass to the cardboard from the inside.



STEP 05 Mark the distance at which the phone is in focus. You need to keep the phone upright, and there are many ways to do this. You can stick a vertical or slightly inclined stand, also made of cardboard. Making your own box will make it easy to adjust the phone inside the box for the perfect focus, so keep the top open before finalising the build. We just folded over a piece of cardboard at the exact dimensions of the phone, so that it fits snugly in the box.



STEP 06 If you fix the holder for the phone on the inside of the box, you can even project on the ceiling. Get some popcorn, dim the lights, and enjoy your new projector. For best results, keep the phone on the highest brightness.